

Family Systems Intelligence in Neurodevelopmental Disability Care: A Conceptual Framework for Family-Centered Service

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Abstract

Systems intelligence (Doherty, 2024; Doherty et al., 2026) offers a pathway for understanding how people perceive and act within complex interconnected systems. This article applies that concept to neurodevelopmental disability (NDD) service systems, where families and professionals coordinate care across clinical, educational, disability, and community contexts. We propose six components of family-centered interventions that build systems intelligence: systemic perception and self-positioning, shared interpretation, role negotiation, coordinated action, feedback responsiveness, and transferable capability. Three examples of family-centered approaches illustrate complementary pathways for developing systems intelligence across family, professional, and service-system relationships. The framework describes systems intelligence as a shared family-service capacity and identifies practical directions for research, training, service design, and equity.

Keywords: family-centered care; family systems; health systems; intervention; systems intelligence; neurodevelopmental disabilities

Introduction

Many families navigate care across multiple service systems. For families of children with neurodevelopmental disabilities (NDDs) or other disability-related and complex care needs, everyday care links caregivers and children with clinicians, educators, social workers, disability service providers, administrators, and community organizations. Fragmented communication, duplicated assessments, unclear roles, inaccessible services, and poor coordination often position

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families as informal system navigators (McCrossin et al., 2022). Fragmentation may persist even when families communicate regularly with each service, because information, expectations, and responsibilities remain poorly integrated across service relationships. Reducing this burden requires services to *recognize* families' experiential expertise—their knowledge of the child, family life, priorities, and everyday contexts—and *integrate* it with professional clinical and service expertise. Care becomes more coherent and meaningful when families, professionals, teams, organizations, and community supports interpret these forms of knowledge together, understand their interdependencies, and coordinate action across settings, while responsibility for making the system work remains distributed rather than transferred to families. This article develops a six-component framework for translating this principle of shared responsibility into the design and delivery of family-centred interventions.

Understanding how families and service partners can share this work requires a concept that connects systemic understanding with coordinated action. Doherty's concept of *systems intelligence* provides that starting point: the capacity to understand how people are connected in multilateral systems and to act constructively within them (Doherty, 2024). We extend this formulation by considering how systems intelligence can develop as a shared capacity and how responsibility for coordinated action can be distributed among families, professionals, teams, and organizations. From this perspective, systems-intelligent action attends to how information moves, how responsibilities are negotiated and aligned, and how relationships shape what is possible in everyday life.

Family systems intelligence is a domain-specific application of the broader construct in which family relationships are central (Doherty et al., 2026). It involves understanding family

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interaction patterns, recognizing how multilateral relationships shape one another, and acting in ways that support personal and relational health. As a developmental capacity, it can be cultivated over time and carried into future family challenges; it may also be expressed intuitively, without formal knowledge of family systems theory.

NDD service systems make the connection between family systems intelligence and broader systems intelligence especially visible. A child's care involves caregivers, siblings, clinicians, educators, therapists, administrators, peer supports, and community services. A recommendation made in one setting can reshape school planning, family routines, employment, transportation, and relationships with the child. Family systems intelligence supports recognition of how care demands affect roles, routines, relationships, and well-being within the family; broader systems intelligence supports shared interpretation and coordinated action across family-service and service-to-service relationships. The development and expression of these capacities must therefore be distributed across the family-service system. Families contribute experiential knowledge and may benefit from language and confidence to describe needs, negotiate roles, and seek support; professionals and organizations are responsible for recognizing family context, power differences, and coordination burden and for designing service processes that do not shift additional work onto caregivers.

Family-centred service offers an established approach for organizing care around these family and service-system interdependencies. It is grounded in the recognition that families possess essential knowledge about their children, differ in their priorities and circumstances, and provide the relational and community context in which children develop (Kuo et al., 2012; McCarthy & Guerin, 2022; Rosenbaum, 2026; Seniwati et al., 2023; Trute & Hiebert-Murphy,

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2013). It is expressed through an ongoing, individualized process of respectful partnership and shared decision-making (Rosenbaum, 2026). These principles are enacted through relational practices such as eliciting and trusting caregivers' accounts, considering how recommendations affect the whole family, checking how families understand and will communicate clinical information, inviting trusted members of the family's support network into conversations when desired, and arranging follow-up so questions and plans can evolve (Rosenbaum, 2026). Implementation can be constrained by limited staffing and space, heavy workloads, restricted family access, inconsistent communication and training, provider attitudes, family stress, and cultural mismatch (Aljawad et al., 2025). The broader evidence base also remains conceptually and contextually uneven, with inconsistent definitions and limited representation of community settings, culturally and linguistically diverse families, and family members beyond mothers (Butler et al., 2025). Caregivers continue to call for tailored, coordinated, accessible, emotionally responsive care grounded in two-way communication (Pozniak et al., 2024).

Together, family systems intelligence, broader systems intelligence, and family-centred service support a view of systemic understanding and coordinated action as a shared family-service capacity. Interventions can develop this capacity across individuals and multilateral relationships, with responsibility distributed throughout the family-service system. Family systems intelligence focuses attention on relationships and adaptive processes within families; broader systems intelligence extends attention to the reciprocal relationships among families, professionals, teams, organizations, and community supports; and family-centred service provides an applied, value-driven context for bringing these capacities into practice.

To specify what capacities such interventions need to develop, this article proposes six components: systemic perception and self-positioning, shared interpretation, role negotiation, coordinated action, feedback responsiveness, and transferable capability. The Family Check-Up[®] (FCU[®]), ENVISAGE, and caregiver peer support illustrate distinct pathways for developing these capacities from different positions in the family-service system, with implications for research and service design.

Six Components of Interventions That Build Systems Intelligence

In this framework, *systems intelligence* refers broadly to the shared capacity of families, professionals, teams, and organizations to perceive, interpret, and act within multilateral relationships (Doherty, 2024). This broader capacity encompasses *family systems intelligence* when attention turns to relationships and adaptive processes within the family, while also extending to relationships between families and the professionals, organizations, and community supports around them (Doherty et al., 2026). Interventions may therefore build family systems intelligence, broader systems intelligence, or both. They do so by helping members of family-service systems notice patterns across family and service contexts, form shared interpretations of needs and responsibilities, negotiate roles and power, coordinate action across settings, respond to feedback, and carry learning into future encounters. Different interventions may develop these components through different relationships and at different points of need, while responsibility remains justly distributed across families and service systems.

The framework comprises six connected components (Figure 1). **Systemic perception and self-positioning** helps members of the family-service system identify relevant people and groups, relationships, constraints, feedback mechanisms, and their own influence within the

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system. **Shared interpretation** creates structured opportunities to align understandings of needs, priorities, and expectations. **Role negotiation** creates explicit opportunities to clarify caregiver and professional expertise, decision-making responsibility, and the work expected of those involved in care. **Coordinated action** connects family goals and capacities with clinical recommendations, service supports, and everyday routines. **Feedback responsiveness** enables families, professionals, teams, and service settings to learn from how plans work in everyday implementation and to revise interpretations, roles, actions, and service responses when circumstances or outcomes indicate that change is needed. **Transferable capability** develops knowledge, relationships, processes, and organizational learning that families, professionals, services, and community partners can carry into future encounters.

Systems intelligence must also be understood within its structural context. Family-centered service occurs through direct encounters, but those encounters are shaped by how services are resourced, organized, and made accessible, as well as by professional hierarchies and broader cultural and societal contexts (Aljawad et al., 2025; McDowell, 2026). These conditions may influence both how work and burden are distributed and whether caregiver expertise is recognized and integrated into processes of care. An intervention that builds systems intelligence therefore attends both to the relationships through which care is negotiated and coordinated and to the broader conditions that enable or constrain those relationships.

This attention to relationships and the conditions surrounding them is consistent with the family systems theory premise that family functioning cannot be fully understood by examining individual members separately; attention must also be given to family structure, organization, and transactional patterns (Miller et al., 2000). Family resilience—how families adapt and thrive

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amid challenges—likewise develops through the interaction of family adaptive processes with social determinants, systemic barriers, and environmental supports (McDowell, 2026). Extending these premises, neither internal family relationships nor family-service relationships can be understood apart from the organizational, cultural, and societal contexts in which they are embedded. We therefore conceptualize systems intelligence as a shared capacity that develops through interaction while remaining shaped by these broader contexts. Communication, collaborative interpretation, role negotiation, and care coordination allow families, professionals, teams, and organizations to develop and exercise this capacity together within the opportunities and constraints those contexts create.

Because family-service systems differ in their relationships, resources, and constraints, shared capacity may develop through different pathways. General systems theory concepts of equifinality and multifinality help explain this variation. Equifinality suggests that coordinated family-centered action may be reached through different pathways, while multifinality cautions that similar interventions or starting conditions may produce different trajectories depending on family circumstances, relational histories, authority, resources, and feedback processes (Bertalanffy, 1968; Shaw et al., 2024). Leadership may therefore shift among families, clinical teams, and peer or community organizations according to the knowledge and coordination a particular task requires.

By locating observable capacities and practices within broader structures and accounting for flexible, dynamic roles, the six-component framework makes family-centered commitments more actionable. The three examples that follow illustrate how these components may be developed through different relationships and intervention pathways (Table 1). They are

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illustrative rather than comparative: each foregrounds a different approach to the six components and represents one possible pathway rather than a complete or equivalent implementation of the framework. Together, the examples show systems intelligence developing through family-engaged assessment and reciprocal feedback that bring multiple perspectives into shared interpretation and coordinated planning; shared family-professional learning that builds common language, role clarity, and transferable capability; and peer relationships that affirm family expertise, broaden systemic perception, and carry practical knowledge into future encounters.

Family Check-Up® as an Intervention That Builds Systems Intelligence Through Ecological Assessment, Feedback, and Planning

The Family Check-Up® (FCU®) is a brief, structured intervention that connects ecological assessment with collaborative feedback and planning (Dishion & Stormshak, 2007). An initial interview elicits caregiver goals, concerns, and motivation. A multi-informant assessment then brings together caregiver and child reports, teacher information, and observations of family interaction. In a feedback session, a trained provider uses motivational interviewing to discuss family strengths, explore caregiver priorities and readiness for change, and collaboratively select appropriate supports. Depending on the family's priorities, options may include parenting modules, community supports, navigation or advocacy assistance, or no further intervention.

This assessment-feedback-planning process fosters the development of systems intelligence. **Systemic perception and self-positioning** begin with an ecological assessment that places child concerns within family relationships, caregiver well-being, school functioning, sociocultural context, and practical constraints. **Shared interpretation** develops as caregivers and professionals consider what the different sources of information from the assessment mean

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in relation to family goals and everyday life. **Role negotiation** occurs through discussion of caregiver priorities, professional recommendations, shared decision-making, and the responsibilities attached to selected supports. **Coordinated action** connects the resulting plan with family routines, services, and community resources. **Feedback responsiveness** is built into the motivational feedback process: caregivers and professionals learn from one another as assessment findings, caregiver responses, priorities, professional knowledge, and readiness for change shape their shared interpretation and evolving plan. **Transferable capability** can develop through the motivation, problem-solving strategies, parenting supports, and system knowledge that families and professionals carry into later challenges.

The FCU[®] has been applied across different service contexts. School-based applications connect family routines and priorities with school concerns and planning (Stormshak et al., 2010), while FCU[®] 4 Health adapts the process for pediatric primary care (Smith et al., 2018). Although the FCU[®] was not developed specifically for NDD services, it has also been implemented and evaluated within a publicly funded Canadian NDD service context, where specialized therapists deliver the model to families of children and youth with complex care needs (Bennett et al., 2024).

ENVISAGE as an Intervention That Builds Systems Intelligence Through Shared Language and Family-Professional Learning

ENabling VISIONs And Growing Expectations (ENVISAGE) is a family-centered, strengths-based program co-designed by caregivers, service providers, and researchers for families of children with NDDs. ENVISAGE-Families consists of five workshops that support caregivers in understanding disability and child development within family and community life, attending to

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caregiver well-being, and developing strategies for communication, collaboration, and connection with family members and service providers (Miller et al., 2022; Miller et al., 2023). ENVISAGE-Service Providers (ENVISAGE-SP) adapts these concepts for service providers participating in their own five-workshop program, while a recently piloted ENVISAGE-Integrated brings caregivers and service providers together for shared learning (Pozniak et al., 2022; Soper et al., 2026).

These workshops foster all six components of systems intelligence. **Systemic perception and self-positioning** develop as caregivers and service providers consider the child within family and community life and reflect on their own positions in family-service relationships. **Shared interpretation** is supported by strengths-based language through which caregiver experience and professional knowledge can inform a common understanding of disability, development, and well-being. **Role negotiation** occurs as caregiver expertise, professional contributions, and benefits of communication and collaboration are made explicit. **Coordinated action** is supported through proposed strategies for carrying shared understandings into conversations, decisions, and plans across family and service settings. **Feedback responsiveness** is reflected in the program's co-design and integrated learning processes, through which family and professional experience can alter shared tools, interpretations, and subsequent responses. **Transferable capability** develops through communication, collaboration, reflection, and advocacy strategies that caregivers and service providers can carry into future family and service encounters.

ENVISAGE-Families has been found feasible and acceptable, with evidence of increased family empowerment, caregiver confidence, perceived legitimacy as members of the child's

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team, and capacity to communicate and collaborate with healthcare providers, educators, and others (Miller et al., 2023). Importantly, family involvement also shaped the program's content, delivery, measures, interpretation, and dissemination, demonstrating how lived experience can influence both intervention participation and program development (Pozniak et al., 2022). Across its family, provider, and integrated forms, ENVISAGE creates related learning opportunities from different positions within the family-service system.

**Caregiver Peer Support as an Intervention That Builds Systems Intelligence Through
Lived Expertise and Community Connection**

Caregiver peer support is a family-led intervention that connects families with other caregivers who bring lived experience of disability, service systems, and family life. The Family Support Institute of BC's Resource Parent/Peer model provides one example: families of people with disabilities connect with peer volunteers who offer emotional support, practical knowledge, and community connection (McCrossin & Lach, 2023). Through reciprocal relationships and storytelling, peer supporters help families interpret service pathways, identify useful questions, anticipate transitions, and locate resources that fit their circumstances. Peers may also accompany families to appointments, helping them bring family expertise and priorities into NDD service encounters.

Systemic perception and self-positioning develop as peer supporters draw on previous encounters to explain how services are organized, where decisions are made, which patterns recur, and how families are positioned within these processes. **Shared interpretation** emerges through storytelling and validation that help families understand their experiences relationally and contextually and develop the vocabulary needed to bring their expertise, needs, and best

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hopes into service conversations. **Role negotiation** is supported as caregivers recognize the value of their lived expertise, clarify the contributions they can make, and negotiate responsibilities with professionals and services. **Coordinated action** draws on the peer network's collective knowledge of which resources are meaningful in practice and how families can request and access them, connecting family priorities with referrals, advocacy, and transition planning. **Feedback responsiveness** occurs as reciprocal peer relationships adapt to families' experiences and peer networks communicate which information, support, and coordination practices communities find trustworthy, accessible, and culturally meaningful, providing NDD service settings with knowledge for ongoing improvement. **Transferable capability** develops as families carry greater system knowledge, confidence, connection, and advocacy capacity into future encounters. In some cases, those who initially receive support later become peer supporters themselves, finding meaning in sharing their experience and extending this knowledge to other families (McCrossin & Lach, 2023).

Peer support develops systems intelligence within families and community networks, while formal services remain responsible for delivering coordinated, accessible, and responsive care. Peer networks can identify the information, support, and coordination practices that communities consider trustworthy, accessible, and culturally meaningful. Integrating these insights can help NDD services better respond to groups whose experiences have been underrepresented in mainstream models of care (McCrossin et al., 2025).

Implications and Remaining Questions

Building on Doherty's formulations of systems intelligence (Doherty, 2024) and family systems intelligence (Doherty et al., 2026), this article extends these concepts to family-centered service

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in NDD contexts and makes three related contributions. First, it extends attention from how individuals exercise systems intelligence within family and medical relationships to how systems intelligence may also develop as a shared capacity across families, professionals, teams, organizations, and community networks. Second, it specifies six components through which interventions can build it: systemic perception and self-positioning, shared interpretation, role negotiation, coordinated action, feedback responsiveness, and transferable capability. Third, it uses the FCU®, ENVISAGE, and caregiver peer support to demonstrate how these components can be developed through different relationships and intervention pathways within the opportunities and constraints created by broader service, cultural, and societal contexts.

The six-component framework is not intended as a standalone theory of how families adapt and thrive in the context of NDD services. Rather, its strengths-based emphasis and focus on adaptation amid adversity align with family resilience scholarship. Walsh's developmental systems framework organizes family resilience around belief systems, organizational processes, and communication and problem-solving processes (Walsh, 2016, 2021). Meaning-making and communication align with systemic perception and shared interpretation; flexible organization and collaborative problem solving align with role negotiation, coordinated action, and feedback responsiveness. Transferable capability similarly reflects adaptive capacities carried into later challenges. Our framework extends this strengths-based focus into family-service relationships, where families, professionals, teams, and NDD service settings can also learn and adapt together.

This strengths-based orientation is necessarily contextual. From a social-ecological perspective, resilience depends not only on families' capacity to navigate and negotiate for resources, but also on whether culturally and contextually meaningful resources are available and

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accessible (Ungar, 2010). McDowell's Contextual Family Resilience Framework further identifies pathways through which social determinants shape family adaptation: inequities in health care, education, and economic stability can amplify stress; access to coordinated care, safe housing, consistent income, and social support can enable families to reorganize roles, sustain communication, and make meaning; and equity-informed interventions can support adaptation by addressing discrimination and other systemic barriers (McDowell, 2026). These perspectives show why interventions that build systems intelligence cannot be reduced to strengthening families or improving professional expertise in isolation. They must develop the relational and service capacity through which family experiential expertise and professional clinical and service expertise can be integrated, responsibilities negotiated, and coordinated action sustained, while addressing the social and material conditions that enable families and professionals to exercise it.

The framework also clarifies what family-centered service must enable within and between families, professionals, and NDD service settings. Families bring experiential expertise, priorities, best hopes, and knowledge of everyday life; professionals bring clinical and service expertise. Responsibilities should reflect relevant knowledge, role, resources, decision-making power, and practical capacity to act, including professionals' and organizations' greater capacity to modify many service processes. Systems intelligence should not depend on individual caregivers or exceptionally skilled professionals. Family-service relationships need protected time and processes through which these forms of knowledge can be interpreted together, roles negotiated, and plans revised through reciprocal feedback. NDD service settings must provide structures that value and sustain shared understanding, coordination, and responsiveness across

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encounters and settings, while distributing coordination and decision-making work without placing additional demands on families.

The three examples illustrate flexible relational pathways rather than a single model for building systems intelligence. The FCU[®] organizes family-engaged assessment, reciprocal feedback, and collaborative planning; ENVISAGE develops shared language through family and professional learning; and caregiver peer support mobilizes lived expertise and community connection. Consistent with equifinality, different pathways may support coordinated family-centered action. Consistent with multifinality, the same intervention may lead to different trajectories as family priorities, relationships, resources, and service conditions vary. Family-service relationships must therefore adapt the balance of assessment, learning, peer connection, professional support, and leadership to the needs and capacities present, while keeping responsibility for coherence and coordination distributed across the system.

Several questions remain. Research can examine whether interventions that build systems intelligence reduce coordination burden, clarify roles, strengthen family-professional relationships, and make NDD services more responsive. It should also examine whether these interventions strengthen the capacity and well-being of professionals providing care and contribute to child and family health, participation, and well-being. Existing measures capture parts of this framework. The Measure of Processes of Care 2.0, for example, assesses caregivers' experiences of communication, service availability, attention to family well-being, and care coordination (King et al., 2025). Although these domains overlap with several components of the framework, caregiver-reported service experience does not by itself assess systems intelligence

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as a shared capacity. Research will therefore need to incorporate family, professional, relational, and service-setting perspectives.

Future research should examine how capacity-building can extend beyond professional training. Studies should continue to evaluate approaches that help families articulate and mobilize their experiential expertise, communicate their priorities and best hopes, and exercise greater influence within service relationships. Research should also examine how professionals, teams, and NDD service settings can better recognize this expertise, integrate it with clinical and service knowledge, negotiate responsibility, and respond to feedback. Implementation research can identify the conditions that support these reciprocal processes, how contributions and leadership shift according to need and context, and how interventions accommodate different family-service trajectories. Equity should remain central to this agenda, including whether interventions expand families' influence while reducing, rather than reproducing, burdens created by inaccessible, culturally mismatched, or discriminatory service systems.

Conclusion

Viewed through a systems-intelligence lens, family-centered service is not only expressed in individual encounters; it is also shaped through relationships and the service contexts that support or constrain them. The six components offer a common language for examining how family and professional expertise inform shared understanding, responsibility, and coordinated action without prescribing a single intervention pathway. The next task is to examine whether this framework can guide interventions that expand families' influence, strengthen reciprocal learning, and reduce coordination burden across diverse cultural and service contexts. Its

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practical value will lie in helping families, professionals, and service systems share responsibility for care while keeping family priorities and everyday life at the center.

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Tables and Figures

Table 1. How three family-centered approaches enact the six components of interventions that build systems intelligence

Component	Family Check-Up [®]	ENVISAGE	Caregiver peer support
Systemic perception and self-positioning	Uses ecological, multi-informant assessment to locate family concerns within relational, school, and community contexts.	Situates the child within family and community life and helps caregivers consider their position within service relationships.	Makes service pathways, useful resources, recurring patterns, and families' positions within systems more visible through lived experience.
Shared interpretation	Uses collaborative feedback to interpret assessment findings alongside caregiver goals, strengths, and concerns.	Develops strengths-based language through facilitated reflection and shared family-professional learning.	Uses storytelling, validation, and experiential knowledge to interpret family needs and service experiences.
Role negotiation	Elicits caregiver priorities and uses collaborative goal-setting to clarify decisions and next steps.	Affirms caregiver expertise and supports clearer roles in communication and collaboration with providers.	Recognizes lived expertise and supports families in articulating needs, asking questions, and negotiating with services.
Coordinated action	Connects assessment and family priorities to tailored supports, referrals, and action plans.	Provides communication and collaboration strategies that can connect action across family and service settings.	Links families with practical resources and supports navigation, advocacy, and planning across services.
Feedback responsiveness	Uses reciprocal assessment and feedback so caregiver responses and professional knowledge shape shared interpretations and evolving plans.	Embeds reciprocal learning through co-design, co-facilitation, program revision, and integrated family-professional learning.	Adapts support through reciprocal peer relationships and communicates community knowledge about trustworthy, accessible, and culturally meaningful practices to NDD service settings.
Transferable capability	Develops motivation, problem-solving, and strategies that can support later family and service challenges.	Builds communication, collaboration, confidence, and advocacy capacities for future encounters.	Builds practical system knowledge, confidence, connection, and advocacy capacity that families can carry into transitions.

NDD CARE

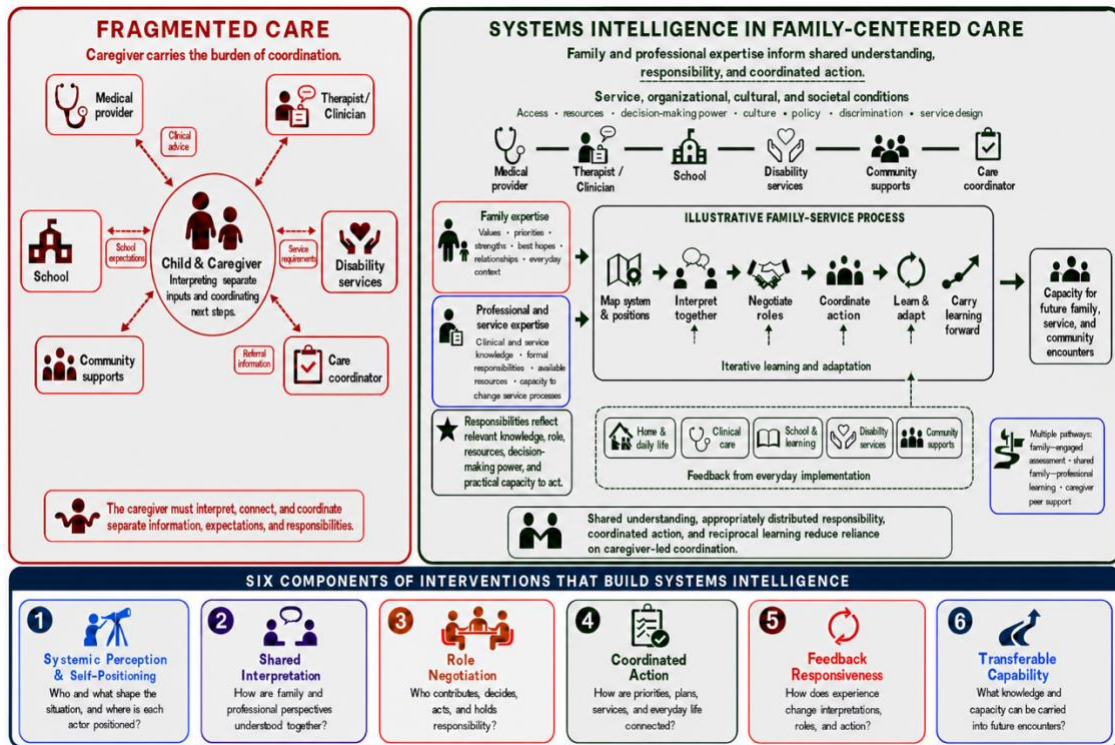


Figure 1. Fragmented care and systems intelligence in family-centered care, including the six components of interventions that build systems intelligence.